Liv DesRoches

603-703-5135 | liv@desroches.org | linkedin.com/in/desrochesl | github.com/desrochesl

Education

Wentworth Institute of Technology Boston, MA

Bachelor of Science in Data Science, Minor in Applied Mathematics *Sep. 2024 – Aug. 2027*

Lee Scholar — GPA 3.8/4.0 — Dean's List (Fall 2024-Spring 2026) — Upsilon Pi Epsilon Honor Society

Relevant Coursework: Algorithms, Data Science Fundamentals, Probability and Statistics, Data Structures.

Experience

AI Research Co-op May 2026 – Aug. 2026

Wentworth Institute of Technology *Boston, MA*

- **Superfund RAG Pipeline:** Built automated retrieval-augmented generation (RAG) evaluation pipelines using LLMs to analyze environmental documents, significantly improving response and retrieval quality.
- Integrated Docling to parse complex unstructured PDF layouts into machine-readable formats, processing 300,000+ EPA Superfund site records.
- **Superfund Document Scraper:** Developed automated web scrapers (Playwright, BeautifulSoup4) and PDF extractors to collect environmental documentation, standardizing extracted metadata for downstream NLP pipelines.
- Engineered multi-threaded Python scripts to bypass rate limits and accelerate bulk document retrieval and processing by 20+ times.
- Orchestrated Slurm batch jobs on the Unity High Performance Computing Cluster, scaling workloads to optimize large-scale processing of over 1.8 TB of unstructured data.

Leadership Experience

Assistant Instructor and Event Facilitator 2017 – Present

Buzz Durkin's Karate School *Atkinson, NH*

- Instructed group lessons for 15+ students, leveraging 15+ years of continuous martial arts practice.
- Managed crowds of event attendees, directing them to their events

Sexuality and Gender Alliance Club President 2026 – Present

Wentworth Institute of Technology *Boston, MA*

- Managed organizational budget requests and delivered formal presentations.
- Organized club meetings and events.
- Created a safe and welcoming environment for students to engage with like-minded peers.

Technical Skills

Languages: Java, Python, Rust, C, F#, SQL (MySQL), HTML/CSS, R
Developer Tools: Git, VS Code, Jupyter Notebook, PyCharm, IntelliJ, Eclipse, ArcGIS
Libraries: pandas, NumPy, Matplotlib, PyTorch, Scikit-learn, Playwright, Docling, LangChain, LlamaIndex, Transformers, FAISS